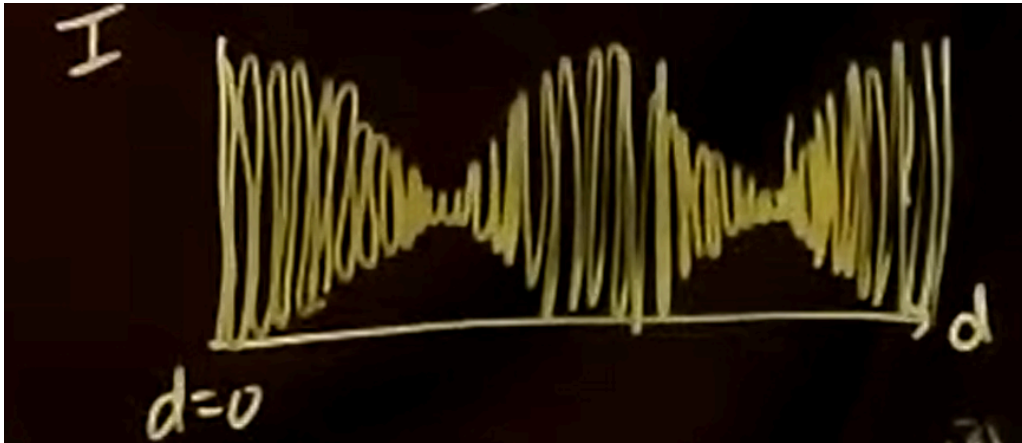


Sodium Doublet Notes

$$I \propto 2 \left[1 + \cos(\omega\tau) \cos\left(\frac{\Delta\omega}{2}\tau\right) \right]$$

where ω is the average frequency of the sodium doublet, and $\Delta\omega$ is the difference



This is a fast cos and a slow cos multiplied together. The slow cos has a frequency $\frac{\Delta\omega}{2}$. If we measure the spacing $\Delta\tau$ between the peaks, we get that

To go from max visibility to minimum visibility, d goes by

$$2\pi \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} \right) d = \frac{\pi}{2}$$

$$d = \frac{1}{\frac{4}{\lambda_1} + \frac{4}{\lambda_2}}$$

If $\lambda_1 = 589.0nm$ and $\lambda_2 = 589.6nm$, then $d = 0.15mm$

We would expect Δd = the spacing between successive maximum visibilities - to be $0.3mm$ for to be the spacing between maximum visibility.